Air University

AIR UNIVERSITY

COAL (Lab)

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Lab Task # 12

Question 01

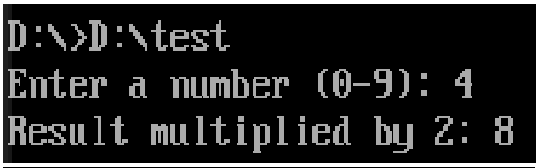
Write an assembly language program to multiply a number by 2 using SHL (Shift Left) instruction.

Code

```
; Single Digit Output
.MODEL SMALL
.STACK 100H

.DATA
    PROMPT     DB 'Enter a number (0-9): $'
    RESULT_MSG DB 0DH, 0AH, 'Result multiplied by 2: $'
    NEWLINE    DB 0DH, 0AH, '$'
    INPUT_NUM  DB ?
    OUTPUT_NUM DW ?

.CODE
MAIN PROC
                   MOV  AX, @DATA
                   MOV  DS, AX

    ; Display prompt
                   MOV  AH, 09H
                   LEA  DX, PROMPT
                   INT  21H

    ; Read single digit input
                   MOV  AH, 01H
                   INT  21H
                   SUB  AL, '0'            ; Convert ASCII to binary
                   MOV  INPUT_NUM, AL      ; Store input

    ; Clear AH for 16-bit operations
                   MOV  AH, 0

    ; Multiply by 2 using SHL (Shift Left)
    ; SHL shifts bits left, effectively multiplying by 2^n
    ; SHL AL, 1 = Multiply by 2^1 = Multiply by 2
                   MOV  AL, INPUT_NUM
                   SHL  AL, 1              ; AL = AL * 2
                   MOV  OUTPUT_NUM, AX     ; Store result

    ; Display result message
                   MOV  AH, 09H
                   LEA  DX, RESULT_MSG
                   INT  21H

    ; Display result
```

```
                   MOV  AX, OUTPUT_NUM
                   CALL DISPLAY_NUMBER

    ; Exit program
                   MOV  AH, 4CH
                   INT  21H
MAIN ENDP

    ; Subroutine to display number (0-18)
DISPLAY_NUMBER PROC
    ; Check if number is two-digit
                   CMP  AL, 10
                   JL   SINGLE_DIGIT

    ; Two-digit number (10-18)
                   MOV  BL, 10             ; Ensure this is 10, not 8
                   DIV  BL                 ; AL = quotient (tens), AH = remainder (ones)

    ; Display tens digit
                   MOV  DL, AL
                   ADD  DL, '0'
                   MOV  AH, 02H
                   INT  21H

    ; Display ones digit
                   MOV  DL, AH
                   ADD  DL, '0'
                   INT  21H
                   RET

    SINGLE_DIGIT:
    ; Single digit number (0-9)
                   ADD  AL, '0'
                   MOV  DL, AL
                   MOV  AH, 02H
                   INT  21H
                   RET
DISPLAY_NUMBER ENDP

END MAIN
```

Output



Explanation:

1. Memory Configuration

The program uses the .SMALL memory model with a 100H (256 byte) stack allocation. The data segment contains prompt strings, a single-byte variable for input storage, and a word-sized variable for the output result, organized for efficient access within the 64KB segment limits.

2. Input Processing

User input is captured through DOS interrupt 21H, function 01H, which reads a single character with echo. The ASCII digit is converted to its binary equivalent by subtracting '0' (30H), enabling arithmetic operations on the numeric value rather than its character representation.

3. Bitwise Multiplication Logic

The core arithmetic operation uses SHL AL, 1 to multiply the input by 2. This left shift operation moves all bits one position left, effectively doubling the binary value. For example, 5 (0101b) becomes 10 (1010b). This approach is computationally more efficient than using the MUL instruction for powers of two.

4. Output Display System

The display subroutine implements conditional logic: for results 0-9, it performs direct ASCII conversion; for results 10-18, it separates digits using division by 10. The DIV instruction with divisor 10 extracts tens and ones digits, which are then converted to ASCII characters for display.

5. DOS Interrupt Utilization

The program integrates multiple DOS interrupt 21H functions: 09H for string display, 01H for character input, 02H for character output, and 4CH for program termination. Each function follows the standard register parameter conventions for DOS programming.

6. Program Execution Flow

Execution begins with segment initialization, followed by user prompt display and input collection. After binary conversion and multiplication, the result is formatted and displayed. The program concludes by returning control to DOS, maintaining proper system resource management throughout its execution.

Question 02

Write an assembly language program to display a two-digit number using SHL (Shift Left) instruction.

Code

```
.MODEL SMALL
.STACK 100h
.DATA
```

```
     num  DB 23

.CODE
MAIN PROC
     ; Init data segment
          MOV AX, @DATA
          MOV DS, AX

     ; Load num to AL
          MOV AL, num       ; Load num to AL
     ; Double via shift
          SHL AL, 1         ; Shift left to double

     ; Extract digits via div by 10
          MOV AH, 0         ; Clear AH for div
          MOV BL, 10        ; Divisor = 10
          DIV BL            ; AL / 10: AL=tens, AH=units

     ; Save units in BH
          MOV BH, AH

     ; Print tens
          ADD AL, 30h       ; To ASCII
          MOV DL, AL
          MOV AH, 02h
          INT 21h

     ; Print units
          MOV AL, BH        ; Load units
          ADD AL, 30h       ; To ASCII
          MOV DL, AL
          MOV AH, 02h
          INT 21h

     ; Exit
          MOV AH, 4Ch
          INT 21h
MAIN ENDP
END MAIN
```

Output

Explanation:

1. Memory Configuration

The program uses the .SMALL memory model with a 100H stack. The data segment contains a single initialized variable num with value 23 (17H in hexadecimal), stored as a byte (DB). This static initialization eliminates user input, focusing purely on arithmetic and display operations.

2. Value Loading and Processing

The number 23 is moved from memory location num into the AL register using MOV AL, num. This direct memory-to-register transfer demonstrates efficient data access patterns in assembly language without intermediate steps.

3. Bitwise Multiplication Implementation

The doubling operation is performed via SHL AL, 1, shifting all bits left by one position. This transforms 23 (00010111b) into 46 (00101110b). The SHL instruction provides faster multiplication by powers of two compared to conventional multiplication instructions.

4. Digit Extraction Methodology

After multiplication, the result (46) undergoes division by 10 to separate decimal digits. DIV BL divides AX (with AH=0) by 10, placing the quotient (4) in AL (tens digit) and remainder (6) in AH (units digit), enabling independent processing of each digit.

5. ASCII Conversion and Output

Each digit is converted to ASCII by adding 30h (48 decimal): tens digit (4 → '4') and units digit (6 → '6'). The DOS interrupt 21h function 02h displays each character sequentially, producing the final output "46" on screen.

6. Program Flow and Termination

The program follows a linear execution path: load, multiply, extract digits, convert to ASCII, display, and terminate. Clean termination via DOS function 4Ch ensures proper return to the operating system without memory leaks or unstable states.

THE END